Database – No SQL – Mongodb

Session 8 –

1. **Connect to your MongoDB server using the mongo shell and create a new database called insta_clone.**

Ans:

Start the shell:

Mongosh

Switch to the database:

use insta_clone

2. **Inside the insta_clone database, use the createUser command to add a user named 'storyAdmin' with the 'dbAdmin' role for managing Instagram Stories data.**

Ans:

```
db.createUser({
    user: "storyAdmin",
    pwd: "StoryAdmin@123",
    roles: [
        {
            role: "dbAdmin",
            db: "insta_clone"
        }
    ]
})
```

3. **Create a read-only user called 'feedViewer' in the insta_clone database who can only view (not modify) posts and comments collections.

Hint: Use the 'read' role when creating the user.**

Ans:

```
db.createUser({
    user: "feedViewer",
    pwd: "FeedViewer@123",
    roles: [
        {
            role: "read",
```

```
            db: "insta_clone"
        }
    ]
})
```

If you specifically want to demonstrate the intended collections, create them first:

```
db.createCollection("posts")
db.createCollection("comments")
```

4. **Simulate authentication: Log in as 'feedViewer' and try to insert a new document into the posts collection. Note the error message and explain why it occurs.

Hint: Use the mongo shell with -u and -p flags.**

Ans:

Exit the current shell:

Exit

Connect as feedViewer:

```
mongosh --username feedViewer --password 'FeedViewer@123' --
authenticationDatabase insta_clone
```

Switch to the database:

use insta_clone

Try inserting a post:

```
db.posts.insertOne({
    username: "testUser",
    caption: "Hello Instagram!"
})
```

5. **Use the db.getUsers() command to list all users in the insta_clone database and describe the roles assigned to each. Then, briefly explain in your own words the difference between 'read' and 'dbAdmin' roles.**

Ans:

Log in with an account that has sufficient privileges, then run:

```
use insta_clone

db.getUsers()
```

You should see users similar to:

```
{
    users: [
        {
            user: "storyAdmin",
            db: "insta_clone",
            roles: [
                {
                    role: "dbAdmin",
                    db: "insta_clone"
                }
            ]
        },
        {
            user: "feedViewer",
            db: "insta_clone",
            roles: [
                {
                    role: "read",
                    db: "insta_clone"
                }
            ]
        }
    ],
    ok: 1
}
```